

2016.0 RANGE ROVER (LG), 303-01

ENGINE - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

LEFT CAMSHAFT (G1509135)

REMOVAL AND INSTALLATION

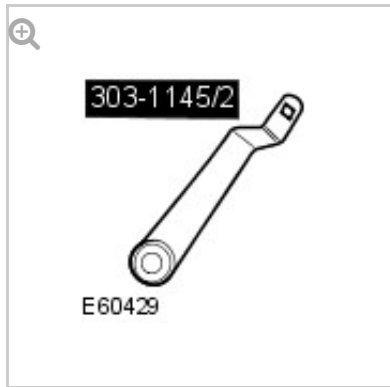
12.13.21	CAMSHAFTS - LH BANK - RENEW	3000 CC, TDV6	9.7	USED WITHINS	+
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SPECIAL TOOL(S)



303-1124

Holding Tool,
Camshaft Front
Pulley



303- 1145/2

Remover, Camshaft
Rear Pulley Bolt



JLR-303- 1523

Remover/Installer,
Camshaft Rear
Pulley

REMOVAL

1. Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).
2. Refer to: [Rear End Accessory Drive](#) (303-05A Accessory Drive - TDV6 3.0L Diesel, Removal and Installation).
3. Refer to: [Timing Belt](#) (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Removal and Installation).

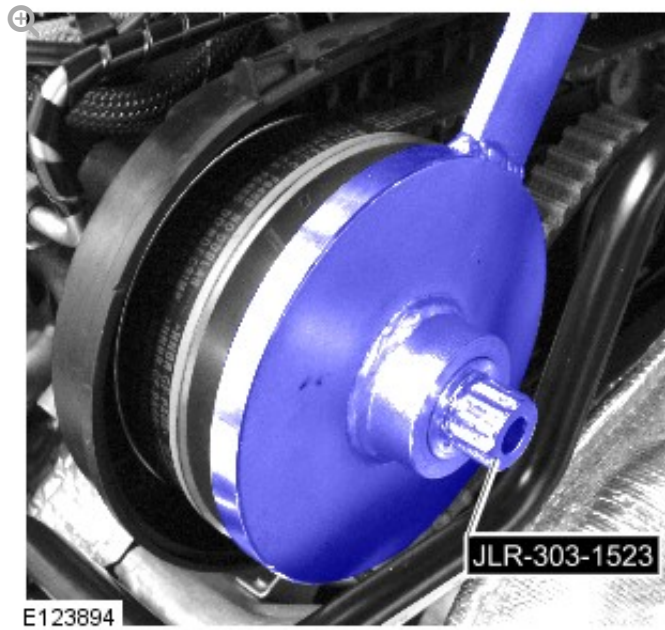
4.



Install the special tool.

Special Tool(s): [JLR-303-1523](#)

5.



Install the special tool.

Special Tool(s): [JLR-303-1523](#)

6.

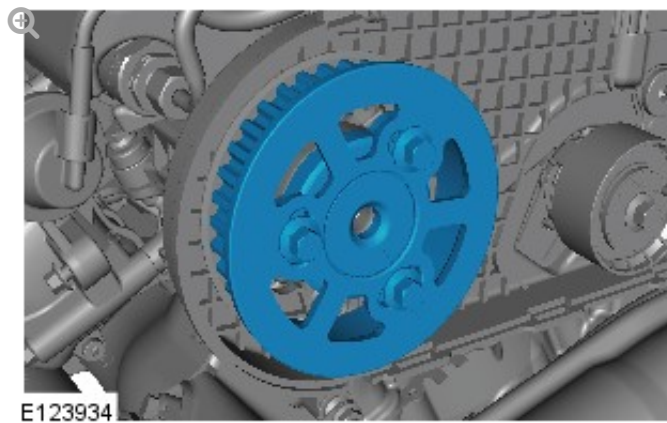


Using the special tool, remove the rear camshaft pulley retaining bolt.

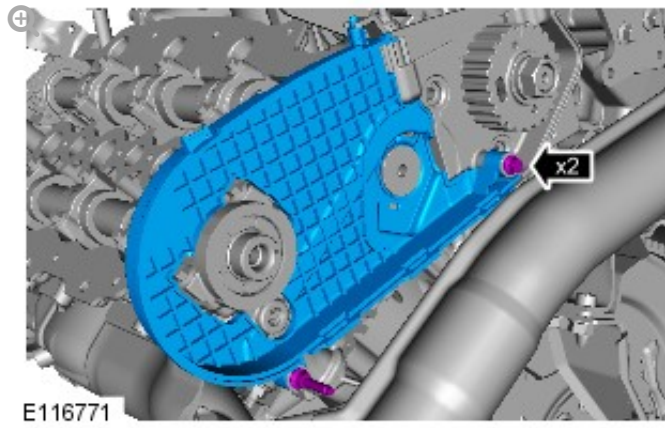
Special Tool(s): [303-1145/2](#)

7. Remove the special tools.

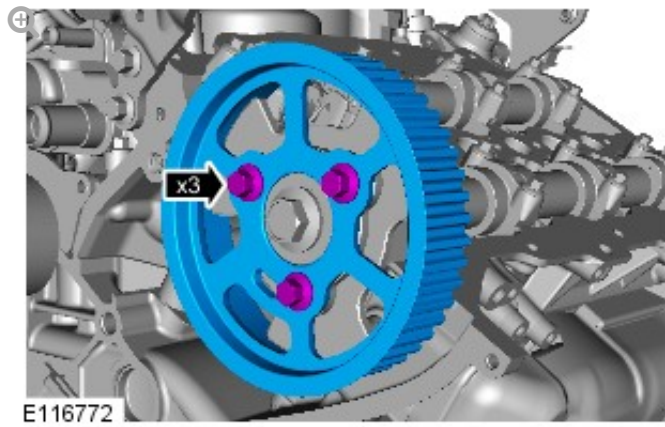
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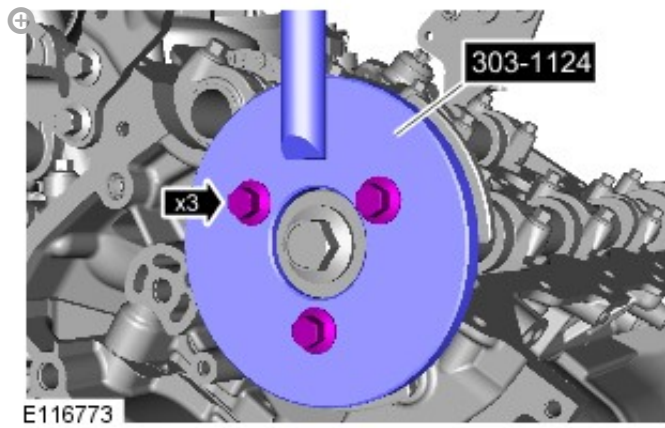
9.



10.



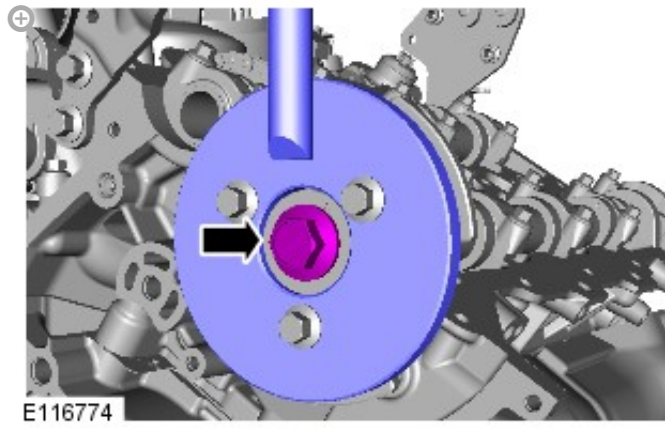
11.



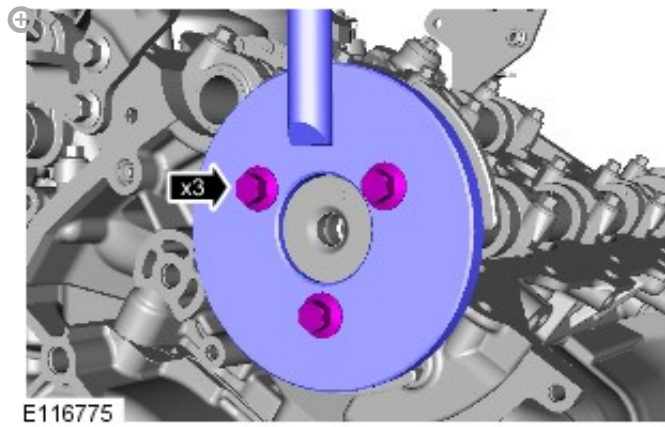
Install the special tool.

Special Tool(s): [303-1124](#)

12.

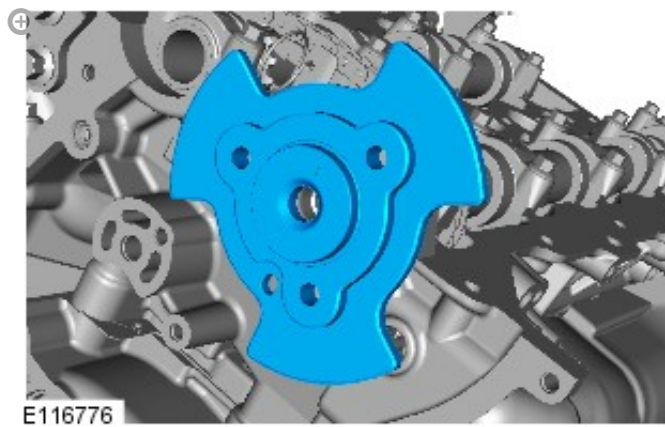


13.

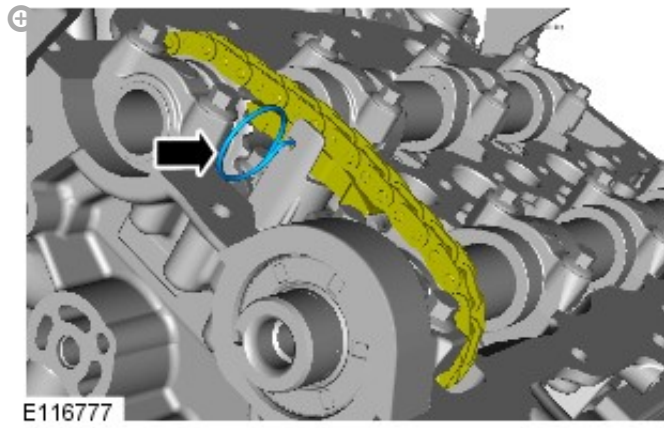


Remove the special tool.

14.

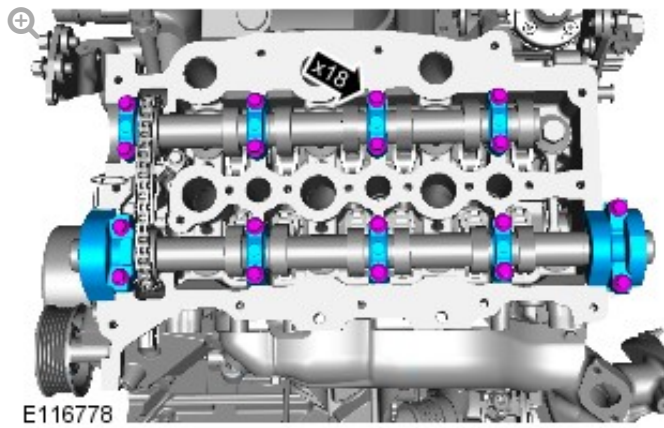


15.



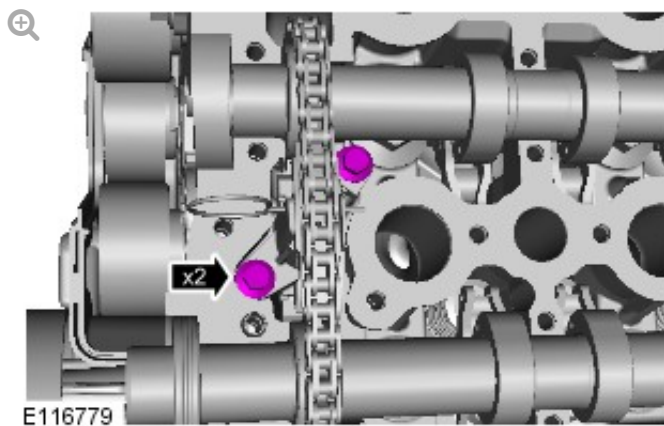
1. Reposition the secondary timing chain tensioner.
2. Retain the secondary timing chain tensioner plunger.

16.



Remove the camshaft bearing caps evenly.

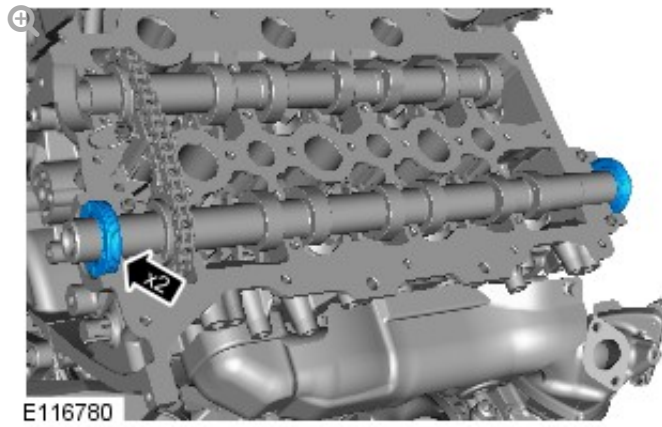
17.



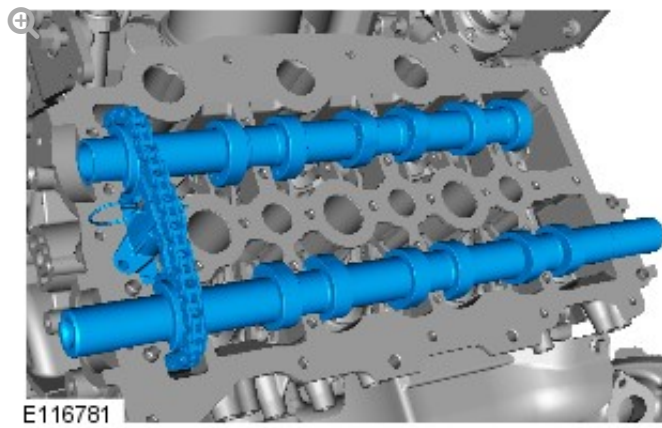
18.

CAUTION:

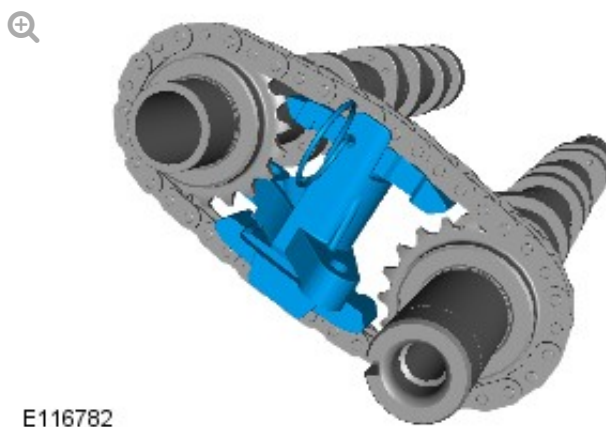
Discard the seals.



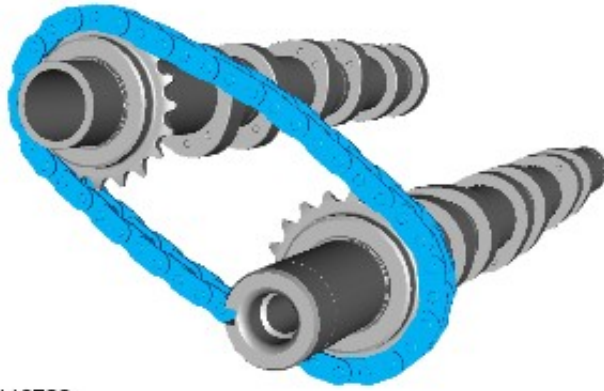
19.



20.



21.

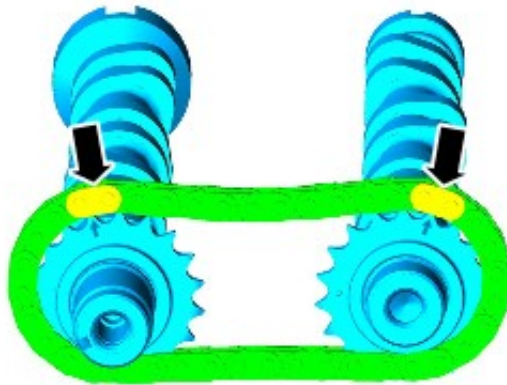


E116783

Remove the secondary timing chain from the camshafts.

INSTALLATION

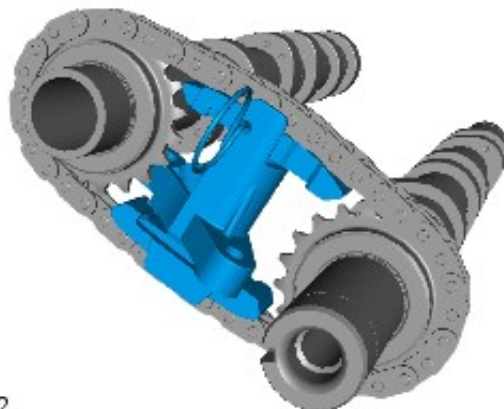
1.



E86461

Install the secondary timing chain onto the camshafts.

2.



E116782

Install the secondary timing chain tensioner assembly.

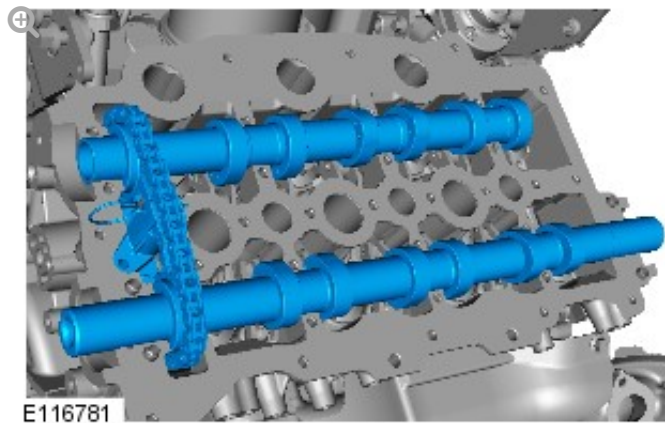
3.

CAUTION:

Make sure that the dots on the camshafts are aligned at the 12 o'clock position. Failure to follow this instruction may result in damage to the engine.

NOTES:

- Lubricate the camshafts and the camshaft bearing caps with oil meeting Jaguar specification prior to installation.
- Use hypoid oil to lubricate the camshafts.

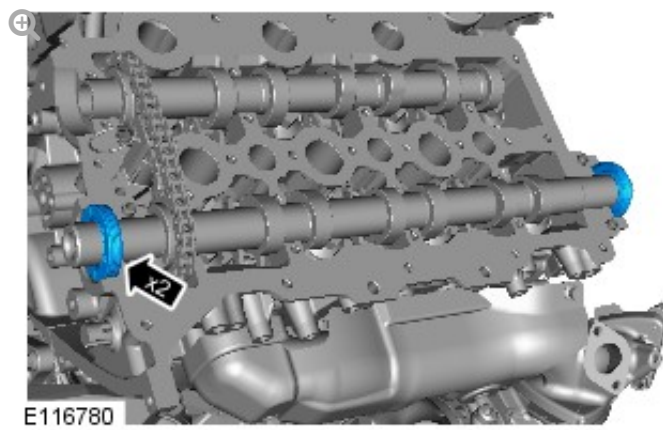


Install the camshafts.

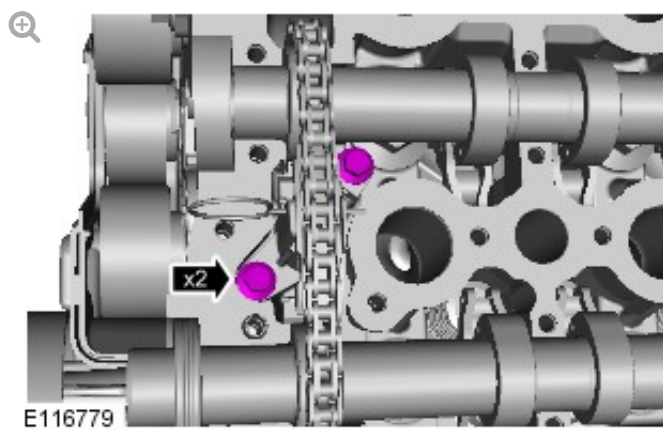
4.

NOTE:

Install new seals.

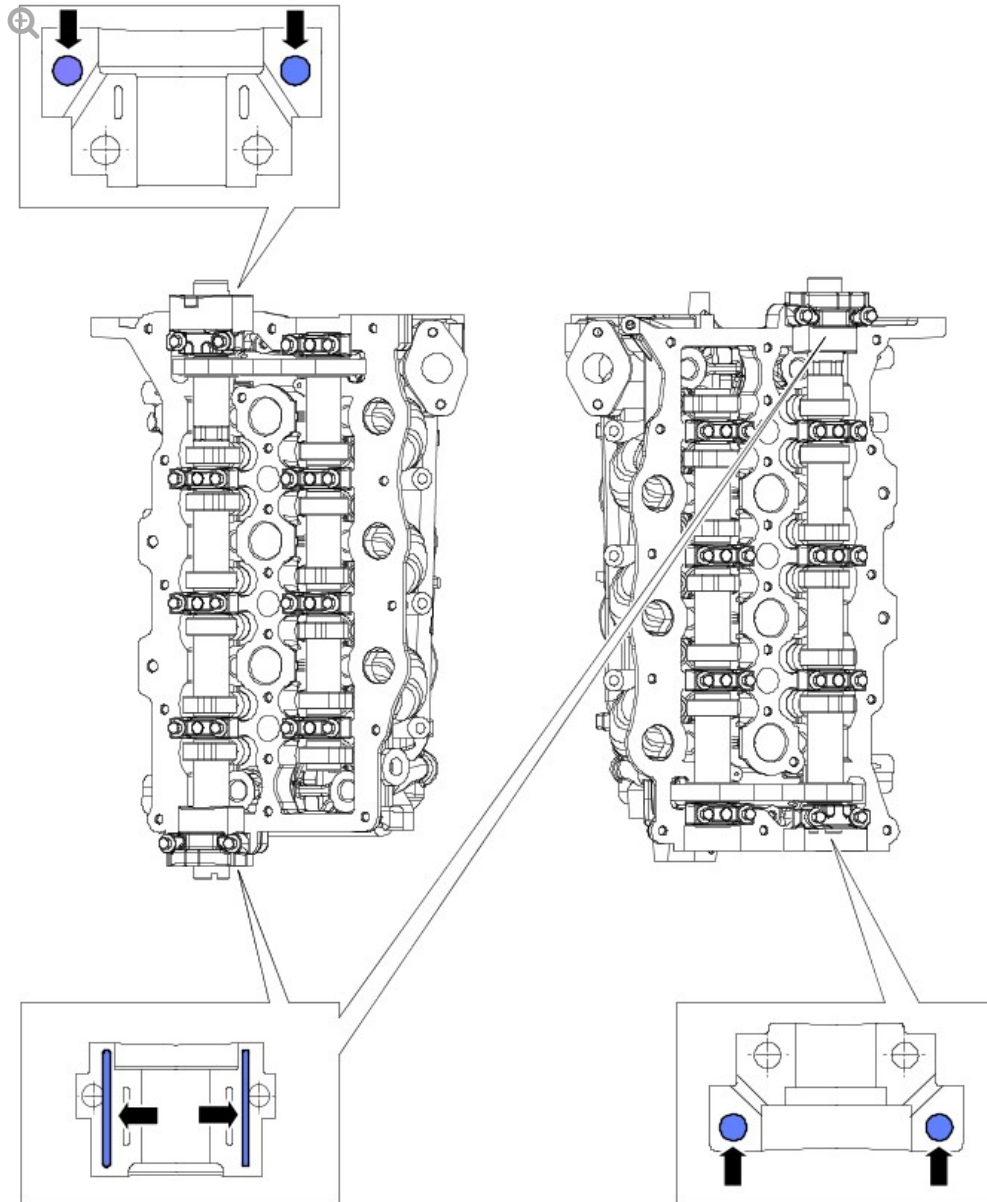


5.



Torque: 10 Nm

6.



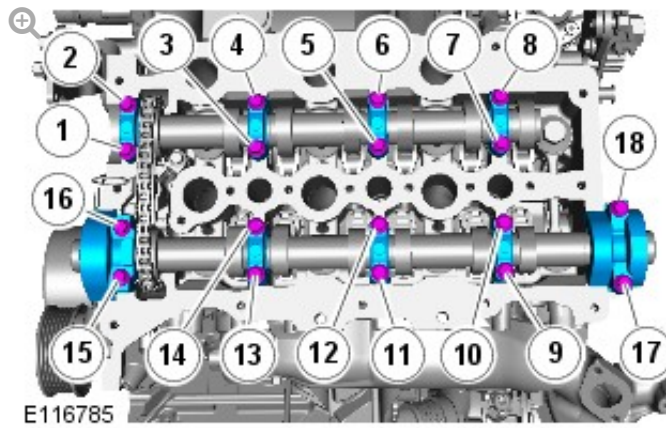
E116784

- Apply Loctite 518 sealant to the exhaust camshaft seal bearing caps.
- Apply Loctite 518 sealant, 2 mm wide, to the LH rear and RH front camshaft bearing caps.
- Apply Loctite 518 sealant, 7 mm diameter, to the LH front and RH rear camshaft bearing caps.

7.

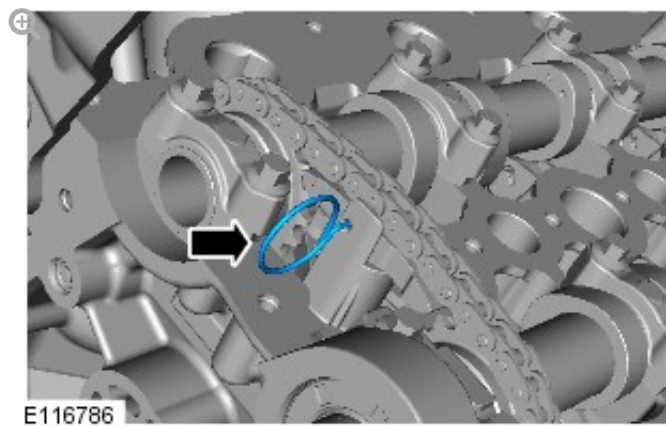
NOTE:

Tighten the bolts in the indicated sequence.



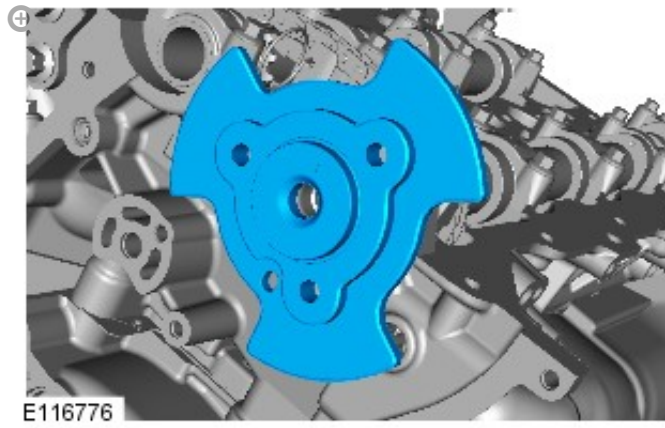
- Install the camshaft bearing caps in their original positions.
- Stage 1: Bolts 1 to 14, 1 Nm.
- Stage 2: Bolts 1 to 14, 5 Nm.
- Stage 3: Bolts 1 to 14, 10 Nm.
- Stage 4: Bolts 15 to 18, 1 Nm.
- Stage 5: Bolts 15 to 18, 5 Nm.
- Stage 6: Bolts 15 to 18, 10 Nm.

8.

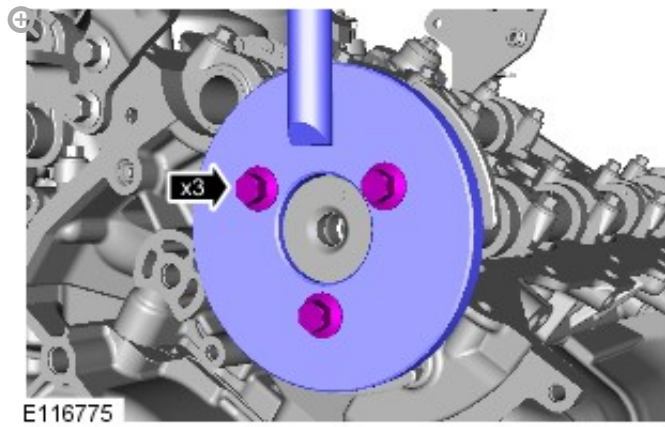


Remove the secondary timing chain tensioner retaining pin.

9.

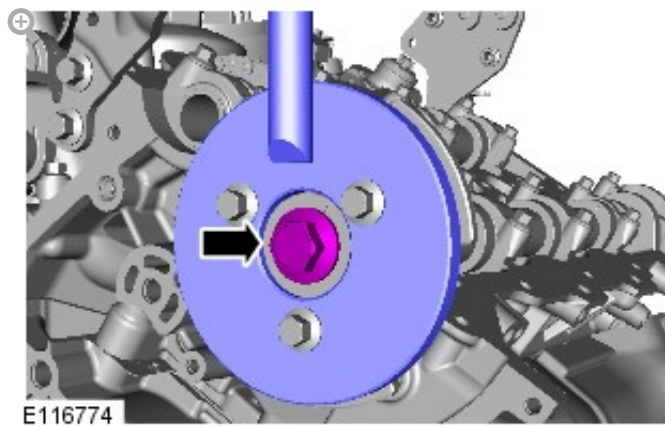


10.



Install the special tool.

11.

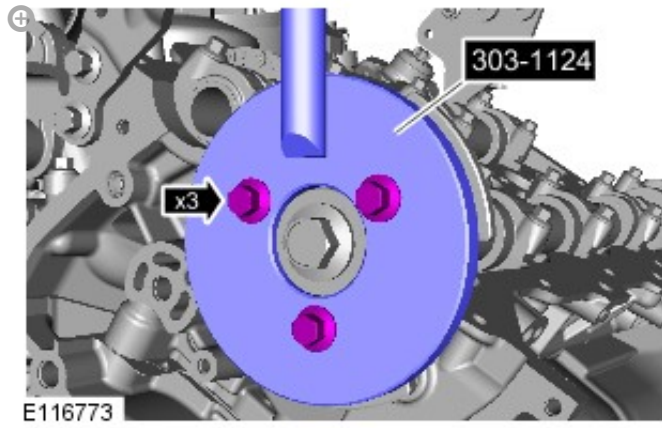


Torque:

Stage 1: 80 Nm

Stage 2: 80°

12.

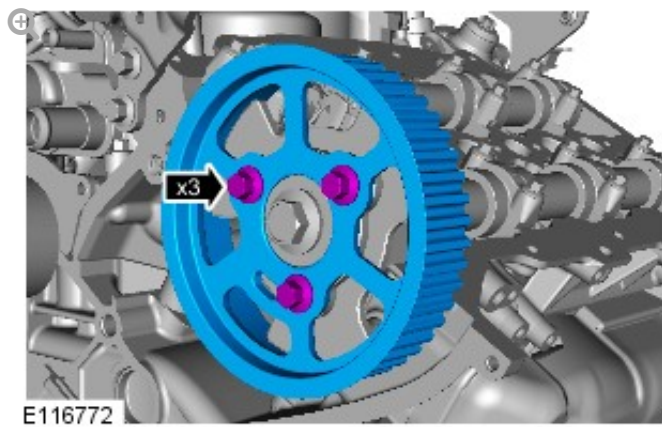


Remove the special tool.

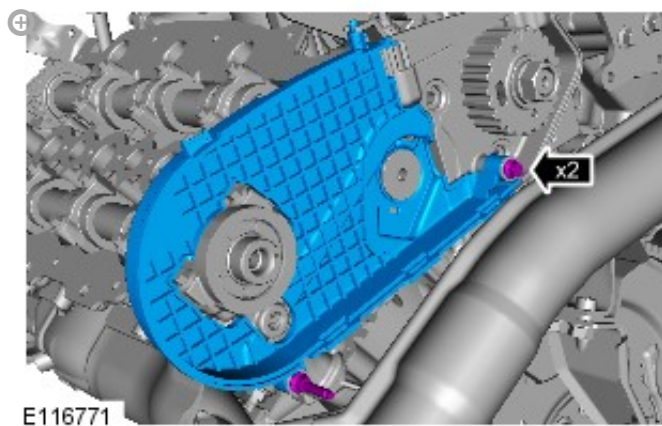
13.

CAUTION:

Only tighten the bolt finger-tight at this stage.



14.

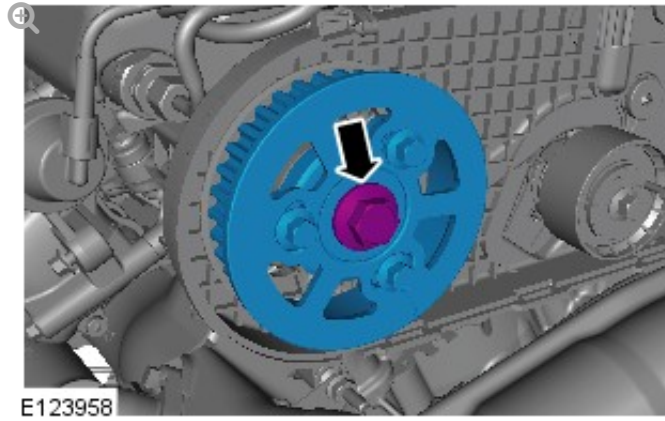


Torque: 10 Nm

15.

CAUTION:

Only tighten the bolt finger-tight at this stage.



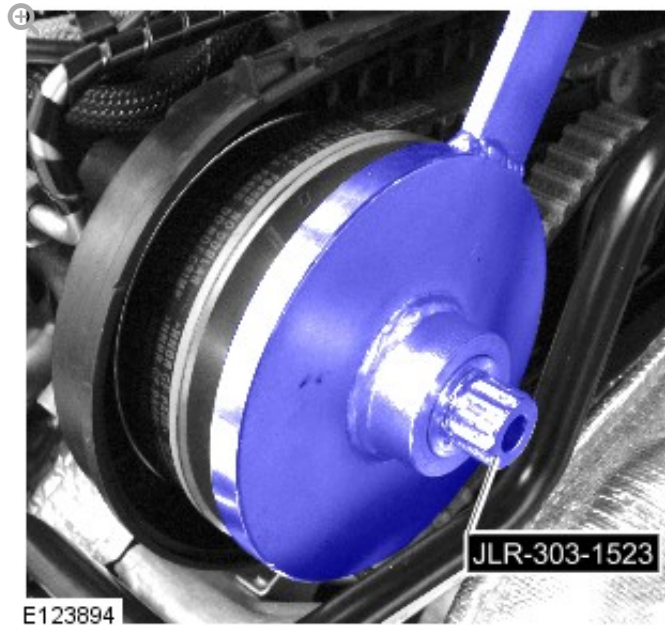
16.



Install the special tool.

Special Tool(s): [JLR-303-1523](#)

17.



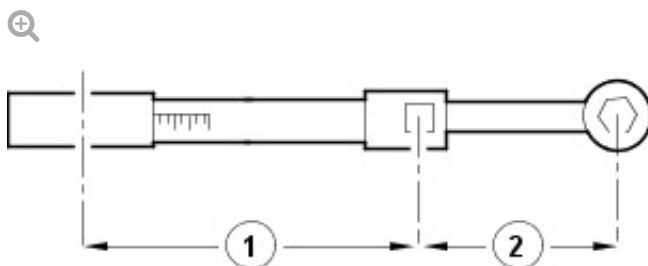
Install the special tool.

Special Tool(s): [JLR-303-1523](#)

18.

CAUTION:

Make sure the torque wrench setting procedure is followed correctly. Failure to follow this instruction may result in damage to the vehicle.



E37107

Calculate the setting for the torque wrench.

19.

CAUTION:

Make sure the torque wrench setting procedure is followed correctly. Failure to follow this instruction may result in damage to the vehicle.



Using the special tool, install the camshaft rear pulley retaining bolt.

Special Tool(s): [303-1145/2](#)

20. Remove the special tools.
21. Refer to: [Timing Belt](#) (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Removal and Installation).
22. Refer to: [Rear End Accessory Drive](#) (303-05A Accessory Drive - TDV6 3.0L Diesel, Removal and Installation).
23. Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

